

# Uber Supply-Demand Gap Analysis

## Exploratory Data Analysis (EDA) Report

### 1. Project Overview

The objective of this project is to analyse Uber ride request data to identify supply-demand gaps and operational inefficiencies. The analysis focuses on understanding ride demand patterns, driver availability, trip completion rates, cancellation behaviour, and location-based supply shortages.

The primary goal is to generate actionable insights that help improve ride completion rates, reduce cancellations, optimize driver allocation, and maximize revenue.

### 2. Dataset Description

The dataset consists of the following key variables:

- **Request ID** – Unique identifier for each ride request
- **Driver ID** – Unique identifier for assigned drivers
- **Pickup Point** – Location of pickup (Airport or City)
- **Request Timestamp** – Date and time when ride was requested
- **Status** – Trip Completed / Cancelled / No Cars Available

From the request timestamp, a new variable **Request Hour** was derived to perform time-based demand analysis.

### 3. Data Preparation and Cleaning

The following preprocessing steps were performed:

- Converted timestamp column into datetime format
- Extracted hour from timestamp for hourly analysis
- Checked for missing values and inconsistencies
- Verified status categories and driver IDs

These steps ensured accuracy and reliability of further analysis.

## 4. Exploratory Data Analysis (EDA)

### 4.1 Hourly Demand Analysis

- Peak demand observed during **6 PM – 9 PM**
- Lowest demand observed during **12 AM – 5 AM**
- Large variation between peak and low hours

**Insight:** Demand is unevenly distributed across the day, indicating supply pressure during evening hours and possible driver idle time during early morning hours.

### 4.2 Trip Status Analysis

Comparison of trip status revealed:

- Significant number of **Trip Completed** rides
- High count of **No Cars Available** during peak periods
- Noticeable **Cancellation rate**, especially in city pickups

**Insight:** Supply shortages during peak hours are leading to unfulfilled ride requests and potential revenue loss.

### 4.3 Airport vs City Analysis

- Airport shows higher instances of **No Cars Available**
- City shows relatively higher **Cancellation counts**

**Insight:** Airport demand spikes may not be adequately supported by available drivers, leading to unmet demand.

### 4.4 Driver Workload Analysis

- Trips per driver were calculated
- Correlation between Request Hour and Trips per Driver was approximately zero

**Insight:** Driver allocation does not dynamically adjust according to hourly demand fluctuations, suggesting potential inefficiencies in distribution.

## 5. Key Business Insights

1. Strong evening demand creates supply shortages.

2. Early morning hours show underutilized driver capacity.
3. High “No Cars Available” cases indicate revenue leakage.
4. Cancellation patterns highlight operational gaps.
5. Driver allocation is not fully aligned with peak demand periods.

## **6. Recommendations**

- Implement surge pricing during high-demand hours.
- Provide targeted incentives for airport pickups during peak times.
- Use predictive demand forecasting for better driver positioning.
- Optimize driver scheduling to reduce idle time during low-demand periods.
- Monitor cancellation trends to improve service reliability.

## **7. Conclusion**

The Exploratory Data Analysis identified clear supply–demand imbalances influenced by time and location. Evening peak hours create pressure on driver availability, while early morning hours reflect lower utilization.

By adopting dynamic pricing strategies, predictive allocation models, and optimized driver scheduling, Uber can improve operational efficiency, enhance customer satisfaction, and increase revenue.